

Semantic Masking for Robust Text Classification Models

Abhijit Suprem¹, Nohelia da Silva²

¹asuprem@gatech.edu; ²nohelia@gatech.edu

School of Computer Science, Georgia Tech

September 18, 2024

1 Final Report

1.1 Semantic Masking

Our proposal centered on Semantic Masking to improve pre-trained models for text classification. Semantic Masking performs targeted masking of specific tokens and words in the input dataset to reduce reliance on strong signals and improve regularization of embeddings in the latent space. We based our proposal on recent recent findings in [2, 1]: for pre-trained classifiers fine-tuned on some dataset and tested on a similar domain dataset different from training dataset, the performance is significantly lower compared to same-dataset accuracy. Our reasoning for semantic masking is as follows: one reason for reduced cross-dataset performance is memorization and overfitting, often caused by overparametrization. We can reduce the chances of overfitting with some token-level augmentations: random token replacement, as well as random token masking. There is a second potential cause of reduced performance: reliance on specific keywords in the input text [1]. If text samples are gathered through keyword search and some type of linear heuristic, that heuristic will be part of the samples themselves, and a classifier may learn that heuristic over actually learning the task. While we might not know the exact keywords and heuristics used in generating a dataset, we can estimate overreliance on keywords with TFIDF and transformer attention.

1.2 Contributions

We planned the following experiments on the fake news datasets in [2] and compare to a baseline without masking. The datasets themselves are 10 public fake news datasets. To this, we planned to add the yearly versions of the NELA dataset [6], which are misinformation headlines in articles from 2018, 2019, 2020, and 2021.

1. Random token masking: We will apply random token masking, with masking probability of 0.15, to train an MLM head along with a classification head.
2. Random word masking: We will apply random word masking, with masking probability of 0.15. While this may mask more tokens than random token masking, we focus

primarily on the word-masking as a way to preserve meaning and context in the masked elements.

3. Semantic masking (word-level) obtained with tfidf: First we apply TFIDF to each of our datasets to obtain more relevant words. Then we mask those words in the dataset and train a classification head.
4. Semantic masking (token-level) obtained with tfidf: We apply TFIDF after tokenizing the dataset(s), then mask the relevant words, and train a classification head.
5. Semantic masking (word-level) obtained with attention: Here, we first use a pre-trained transformer (BERT or ALBERT), with an untrained classification head to rank which tokens the transformer attends to. We then obtain the words corresponding to those tokens, and collect the words that have the k -highest instances of attending by the transformer (we will experiment with $k=50$, empirically, but might perform a sweep across different k values as well). We will then mask these words, and train a classification head.
6. Semantic masking (token-level) obtained with attention: We repeat the above process; in this case, we will use only attending tokens, without collecting corresponding words. The remaining procedure is the same.

2 Progress so far

2.1 Experiments

We first conducted a baseline experiment on the fake news datasets to get the initial same-dataset accuracy and cross-dataset accuracy. We also calculated similar-dataset accuracy. For similar dataset accuracy, we took datasets that have the same content modality and computed cross-dataset within that batch. For example, in Table 1, the similar-dataset accuracy for ‘k.title’ is computed by testing it on ‘miscov’ and ‘c19_title’, since these are both datasets containing article titles.

Our experimental setup is as follows: We use a pre-trained ALBERT classifier [7] that is fine-tuned on each dataset, with a learning rate of $1e-4$. We train each classifier for 5 epochs with early stopping tracking the cross-entropy classification loss. We perform our experiments with PyTorch. We leave tests with BERT for the next stage.

First, a few notes. The CDA is uniformly lower, due to the divergence in training and test data. Even when we reduce that divergence with the SDA tests, there is significant decrease in accuracy. For example, the model trained on ‘k.title’ has accuracy of 0.97 on the k_title test set. On all remaining datasets, it has average accuracy of 0.5, and on datasets with titles (‘miscov’ and ‘c19_title’, the accuracy increases only to 0.6.

2.2 Masking Experiments

We then perform the masking experiments. Currently, we have tested random token masking, random word masking, and top-k word masking using TFIDF. We have attention masking,

Table 1: Baseline accuracies for datasets. Same-dataset accuracy computes the traditional ML accuracy metric. CDA computes cross-domain accuracy on all n-1 datasets. Finally, SDA computes cross-domain accuracy on datasets that are similar in content to the training dataset.

Dataset	Train-Test Splits	Dataset Content	Same Dataset Acc.	Cross-Dataset Acc. (CDA)	Similar-Dataset Acc. (SDA)
k_title	31K/9K	Titles	0.97	0.50	0.60
coaid	5K/1K	Articles	0.97	0.60	0.63
c19_text	2.5K/0.5K	Articles	0.98	0.73	0.61
cq	12.5K/2K	Tweets	0.54	0.53	0.51
miscov	4K/0.6K	Titles	0.55	0.49	0.50
k_text	31k/9K	Articles	0.98	0.56	0.57
rumor	4.5K/1K	Tweets	0.83	0.52	0.55
cov_fn	4K/2K	Tweets	0.96	0.46	0.51
fakeddit	878K/92K	Tweets	0.80	0.49	0.64
nela	699K/158K	Tweets	0.72	0.53	0.60
c19_title	2.5K/0.5K	Titles	0.95	0.49	0.63

plus combinations of masking strategies, remaining.

For random token masking, implement masking following existing code on masked language modeling, where we use the masking input of a HuggingFace transformer to indicate which tokens should be masked. A different set of tokens is masked for each batch in each epoch. We implement random word masking as follows: during tokenized text generation, we track the tokens corresponding to each word. Then, we select a random subset words in the input text to be masked. Finally, we look up corresponding tokens for those words, and mask the tokens.

For targeted masking of specific words, e.g. semantic masking with TFIDF, we first identified the top-k words ranked with TF-IDF on each dataset. Then we preprocessed the dataset to mask those words during training only. We then augmented this with random word masking as well. Our preliminary results are shown in Table 2.

In each case, we are performing a cross-domain experiment, where the Dataset column refers to the training dataset. The testing datasets are the remaining 10 datasets, and we average the accuracies. While this is not a perfect measure of the overall generalizability, due to accuracy discrepancies between datasets, we consider this measure as a first step in evaluating overall generalizability. As we see in Table 2, with random token masking, we are able to slightly improve over the original cross-domain accuracy in most cases. For example, for ‘fakeddit’, the original cross-domain accuracy is 0.49, slightly worse than random guessing. With token masking, we improve to 0.67. With random word masking, we improve slightly over RTM (though in some cases we also decrease accuracy). We attribute this noise to the smaller size of most datasets, and the limited length of the text items. With token masking, 15% of tokens often corresponds to 5 tokens masked on average sentence token length of 35. However, the average length of sentences with respect to words is 15, corresponding to 3 words masked. With 3 tokens per word, this corresponds to 9 masked tokens in random word masking. In smaller datasets, this causes slightly more noise than RTM.

Table 2: Preliminary masking experiments

Dataset	Acc	CDA	SDA	RTM	RWM	TFIDF	CDA Recovery	SDA Recovery
k_title	0.97	0.50	0.60	0.67	0.66	0.67	17%	7%
coaid	0.97	0.60	0.63	0.69	0.77	0.78	20%	16%
c19_text	0.98	0.61	0.61	0.64	0.75	0.69	14%	15%
cq	0.54	0.53	0.51	0.52	0.52	0.53	-1%	4%
miscov	0.55	0.49	0.50	0.51	0.51	0.51	3%	2%
k_text	0.98	0.56	0.57	0.64	0.63	0.70	14%	13%
rumor	0.83	0.52	0.55	0.61	0.59	0.64	15%	11%
cov_fn	0.96	0.46	0.51	0.56	0.59	0.65	19%	14%
fakeddit	0.86	0.49	0.64	0.67	0.75	0.78	33%	16%
nela	0.84	0.53	0.60	0.68	0.68	0.70	21%	12%
c19_title	0.95	0.49	0.63	0.72	0.72	0.79	31%	16%

Finally, with TFIDF (using for key word and random word masking), we improve accuracy better than just RTM or RWM, in most cases. Again, source of noise are due to smaller dataset size. A key example of improvement is on ‘c19_title’, where TF-IDF based masking yields accuracy of 0.79, compared to CDA of 0.49 and SDA of 0.63.

To better compare performance of TFIDF, we calculated accuracy recovery with respect to CDA and SDA. Recovery, in case of CDA, is computed as the difference between accuracy drops between the original accuracy to CDA and the higher accuracy of TF-IDF based masking. That is:

$$Recovery_{CDA} = \left| \frac{Acc - CDA}{Acc} \right| - \left| \frac{Acc - TFIDF}{Acc} \right|$$

For SDA, we simple compare the TD-IDF accuracy to SDA. We see that in most cases, the CDA recovery is 20%, meaning there is 20% *less* accuracy drop between the same-data accuracy and cross-domain accuracy with using masking compared to without. In the case of 2 datasets, ‘cq’ and ‘miscov’, this recovery is low (and in ‘cq’s case, it is actually negative, meaning masking performed worse). This we attribute to the noise in the original dataset itself, since the same-dataset accuracy for these datasets is only slightly better than random guessing (accuracy of 0.54 and 0.55, respectively). The SDA recovery is slightly lower since the SDA accuracies are higher than CDA - meaning the recovery is not as prominent.

2.3 Attention-based masking

We show in Table 3 our results in applying KWM-Att and KWRWM-Att. We have omitted token masking results here since they were not as promising as word-masking results. Keywords selected by attention, along with random attention-based masking performs significant better than without any attention (e.g. CDA accuracy column). We have underlined experiments in the KWM column that did not pass the t-test across our experiments, indicating results where KWM could be better or equivalent to KWRWM due to variance in results.

Table 3: Attention vs TFIDF for masking

Dataset	CDA	RTM	RWM	KWM (TFIDF)	KWRWM (TFIDF)	KWM (Att)	KWRWM (Att)
k_title	0.5	0.67	0.66	0.62	0.67	0.69	0.71
coaid	0.6	0.69	0.77	0.74	0.78	0.78	0.81
c19_text	0.61	0.64	0.75	0.74	0.69	0.78	0.81
cq	0.53	0.52	0.52	0.53	0.53	0.53	0.54
miscov	0.49	0.51	0.51	0.51	0.51	0.54	0.54
k_text	0.56	0.64	0.63	0.61	0.70	0.61	0.62
rumor	0.52	0.61	0.59	0.54	0.64	0.59	0.63
cov_fn	0.46	0.56	0.59	0.56	0.65	0.62	0.65
fakeddit	0.49	0.67	0.75	0.73	0.78	0.79	0.81
nela	0.53	0.68	0.68	0.65	0.70	0.68	0.72
c19_title	0.49	0.72	0.72	0.70	0.79	0.76	0.81

From these preliminary results, it is clear that masking-based pretraining contributes to some generalizability improvement. We note that for some datasets that are well outside remaining datasets, generalizability is almost impossible. This is the case when training with ‘miscov’ or ‘covid_cq’, where regardless of approach, accuracy is around random guessing. While there are some approaches that slightly improve accuracy, e.g. KWRWM-Att with ‘miscov’ improves from 0.49 to 0.54, we attribute this to random training improvements instead of any sustained generalizability improvement. However, the other results are much more promising.

Our next steps are to perform extensive additional experiments on a variety of well-known domain adaptation datasets, e.g. the Amazon Reviewsd dataset consisting of 4 domains, Amazon-50 dataset, and on the NELA yearly subsets. For the latter, we are especially interested in the temporal impact of generalizability, and whether generalizability is temporally scoped or can be extended beyond training time artifacts.

2.4 Detected Words and Tokens

We also show the identified words and tokens from datasets here, as an empirical observation. For the NELA dataset, which contains misinformation tweets, we applied key token and key word detection using TFIDF and our attention-based ranking method.

With TFIDF-tokens, we obtain the following top-5 tokens:

[‘\’, ‘&’, ‘,’ , ‘-’, ‘,’]

With TFIDF-words, we obtain the following top-5 words:

[‘year’, ‘coronavirus’, ‘million’, ‘words’, ‘house’]

With Att-tokens, we obtain the following top-5 tokens:

[‘has’, ‘vaccine’, ‘uk’, ‘could’, ‘be’]

With Att-words, we obtain the following top-5 words:

`['coronavirus', 'covid-19', 'virus', 'pandemic', 'outbreak']`

Empirically, attention yields more interpretable semantic words and tokens relative to TFIDF. We see this impact with higher accuracy in Table 3

2.5 Conclusion

In this report, we have shown semantic masking as a promising technique for improving the generalizability of transformer-based classifiers. By masking strong-signal keywords in text datasets before training, semantic masking helps the classifier to focus on the overall meaning of the text rather than overfitting on individual keywords. As a result, the classifier is better able to generalize to new, unseen examples and to make more accurate predictions. We tested semantic masking on 11 public fake news datasets, and we show promising results, with an average accuracy improvement of 15-20% across a majority of datasets. This improvement suggests that semantic masking can be effective at helping transformer-based classifiers to better identify and classify misinformation. Further research is needed to fully understand the mechanisms behind this improvement and to determine the optimal approach for implementing semantic masking in the context of misinformation detection. Overall, the use of semantic masking has the potential to significantly improve the ability of systems to combat the spread of fake news and to help maintain the integrity of online information.

References

- [1] Moon, Seung Jun and Mo, Sangwoo and Lee, Kimin and Lee, Jaeho and Shin, Jinwoo (2021). Masker: Masked keyword regularization for reliable text classification. *AAAI 2021*.
- [2] Abhijit Suprem and Calton Pu (2022). Evaluating Generalizability of Fine-Tuned Models for Fake News Detection, *CoRR*.
- [3] Wu, Jiajie (2021). Literature review on vulnerability detection using NLP technology.
- [4] Wahle, J. P., Ashok, N., Ruas, T., Meuschke, N., Ghosal, T., and Gipp, B. (2022). Testing the generalization of neural language models for COVID-19 misinformation detection. *International Conference on Information*.
- [5] Lyu, Y., and Tsang, I. W. (2019). Curriculum loss: Robust learning and generalization against label corruption. *CoRR*.
- [6] Maurício Gruppi and Benjamin D. Horne and Sibel Adalı (2021). NELA-GT-2021: A Large Multi-Labelled News Dataset for The Study of Misinformation in News Articles. *CoRR*

- [7] Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., and Soricut, R. (2019). Albert: A lite bert for self-supervised learning of language representations. arXiv preprint arXiv:1909.11942.
- [8] Nakamura, Kai, Sharon Levy, and William Yang Wang. "r/fakeddit: A new multimodal benchmark dataset for fine-grained fake news detection." CoRR (2019).